

RESEARCH ARTICLE

Cozie Singapore: Scalable crowdsourced smartwatch micro-surveys to capture urban heat and noise perception and behavior

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ARTICLE HISTORY

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ABSTRACT

[PLACEHOLDER: 150–250 word structured abstract. Context (urban sustainability and the human experience of everyday space types); the ORENTH / Cozie-Singapore watch-survey dataset (9,962 responses, 106 participants); the aim (characterize how perception, physiology and ambient weather differ across space types — Home, Office, Classroom, Other Indoor, Outdoor, Transportation); the approach (exploratory analysis + effect-size-based statistical characterization + weather augmentation + discriminative modeling); the headline finding (function/behaviour separates spaces, but ambient weather barely marks space while genuinely tracking thermal preference); and the implication for sustainable urban design.]

KEYWORDS

[PLACEHOLDER: 4–6 keywords — e.g. urban sustainability; thermal comfort; wearable sensing; space-type characterization; watch-survey; Singapore]

1. Introduction

A key component of urban studies is the characterization of how the built environment impacts humans. Often, this analysis uses conventional geospatial information from the data-rich mapping context. These geospatial contexts can then be enriched with a significant amount of metadata through convergence with temporal sensor data, imaging data from satellites or street-view imagery, or from smart devices. Street-view imagery has been used to quantify visual and perceived qualities of the streetscape at scale (Ito, Kang, Zhang, Zhang, & Biljecki, 2024), while network-based methods characterize the morphology and connectivity of the urban fabric (Yap, Stouffs, & Biljecki, 2023). Even human presence itself can be inferred indirectly, for example by estimating building occupancy from mobile phone traces (Barbour et al., 2019).

Despite this effort, significant elements of the human experience in the urban context are missing as observation is not the equivalent of capturing perception. Physical measurements alone are weak predictors of how occupants actually judge their surroundings, because non-physical and psychological factors shape perception (Castaldo et al., 2018), and expectations condition the same measured conditions differently (Schweiker, Risetto, & Wagner, 2020). Capturing perception directly has a long methodological

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basis in ecological momentary assessment and experience sampling, which record subjective states in real time and in context (Christensen, Barrett, Bliss-Moreau, & Lebo, 2003; Stone & Shiffman, 1994), and pairing these reports with location data extends them to the geographically explicit case (Kirchner & Shiffman, 2016).

1.1. Crowdsourcing humans-as-a-sensor data across the urban context

Geographically-explicit ecological momentary assessment (GEMA) developed as a set of methods to record momentary subjective states together with the location in which they occur, and it has been applied across public health and urban analysis (Kirchner & Shiffman, 2016). It has been used to link the daily acoustic environment to real-time noise annoyance (Zhang, Zhou, Kwan, Su, & Lu, 2020) and to relate exposure to natural diversity to momentary mental wellbeing (Hammoud et al., 2024). Standardized architectures and reporting guidelines have since emerged to make these studies more comparable and reproducible (Gharani et al., 2021; Kingsbury et al., 2024). The open-source Cozie platform takes this approach a step further by delivering micro-surveys directly on a smartwatch and pairing each response with the device’s onboard physiological and environmental sensors (Tartarini, Frei, Schiavon, Chua, & Miller, 2023). Wearables are increasingly used in this way to capture indoor and environmental quality alongside the people experiencing it (Abboushi et al., 2022), and consumer smartwatches can measure exposures such as ambient noise with useful accuracy (Fischer, Schraivogel, Caversaccio, & Wimmer, 2022).

This study was developed as a follow-on effort to scale up this data collection, with a focus on heat and noise in the urban context. Both exposures are central to the everyday experience of a dense tropical city, where thermal comfort is a persistent concern and noise is a leading source of distraction and dissatisfaction (Appel-Meulenbroek, Steps, Wenmaekers, & Arentze, 2020). Individuals also differ substantially in how they perceive the same conditions, which motivates collecting perception at the level of the person rather than the building (Wang et al., 2018). Linking these reports to geolocated urban context connects them to the digital-twin models increasingly used to represent cities (Ignatius et al., 2024; Miller et al., 2021). The resulting dataset previously supported a public machine-learning competition on predicting heat and noise perception from contextual data (Miller et al., 2023), and this paper instead characterizes the collected experience itself across the range of everyday urban spaces.

2. Methods

This study characterizes human-experience data collected in Singapore with the Cozie platform, an open-source application for the Apple Watch that administers watch-based micro-surveys and records the wearer’s onboard physiological and environmental sensors (Tartarini et al., 2023). The deployment and its instrumentation have been described in earlier methods papers that also report preliminary results (Miller et al., 2023), and the same platform underpins related work on just-in-time interventions for heat and noise (Miller et al., 2025). The design follows established practice for ecological momentary assessment and its geographically-explicit extension, including recommendations for sampling, scheduling, and transparent reporting (Kingsbury et al., 2024; Reichert et al., 2020). Because the platform continuously records location and physiological signals, the deployment also attends to the consent and privacy considerations raised for mobile-sensing studies (Fuller, Shareck, & Stanley, 2017).

The remainder of this section describes the field study and participants, the structure of the micro-survey, the linked wearable, weather, and urban-context data, and the processing steps that produce the survey-level dataset used for the characterization that follows.

2.1. Crowdsourced watch-based micro-survey and physiological and environmental data

The field study reported here is the same deployment described in detail by Miller et al. (2025), and the following is a summary of that protocol, illustrated in Figure 1. Participants were recruited through the National University of Singapore (NUS) Student Work Scheme and by word of mouth, producing a pool made up mostly of NUS students together with a smaller number of adults from outside the university. The cohort was predominantly young, with roughly 73% aged 18 to 30, and comprised 39% male and 61% female participants, and ethical approval was granted by the NUS Institutional Review Board. Each participant used an Apple Watch running the Co-zie application paired with an iPhone, and loan devices were provided to those who lacked a compatible watch or phone. Participants were asked to wear the watch during working hours, from 9 AM to 7 PM on weekdays, for approximately four weeks, and to complete at least 100 micro-surveys together with weekly surveys and an in-person exit interview; responses submitted outside this window were retained. The deployment ran in two phases that differed in how just-in-time intervention messages were triggered. As summarised in Figure 1, each watch-based micro-survey response is time-stamped and geolocated, and is paired with the wearer’s onboard physiological signals and with linked outdoor weather and urban-context data. The present analysis draws on a larger prepared dataset from this deployment, comprising 9,962 micro-survey responses from 106 participants, and focuses on characterizing the reported experience rather than the intervention outcomes.

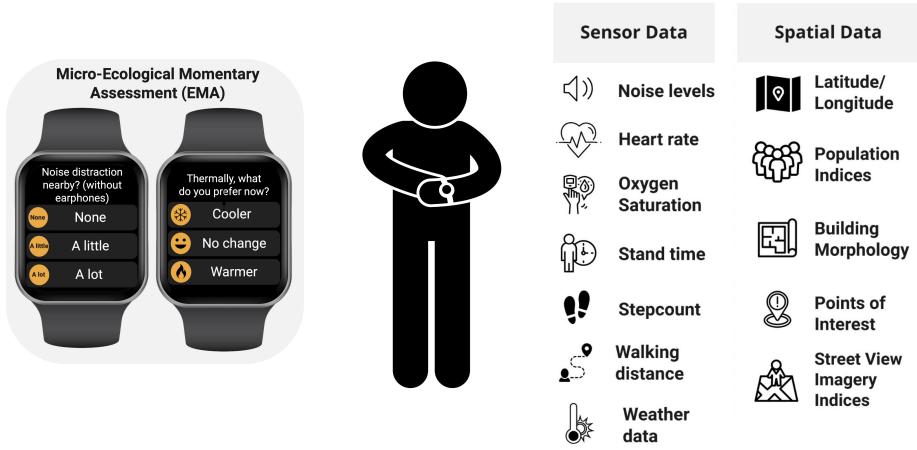


Figure 1. [PLACEHOLDER: Study design and data-collection workflow.]

[PLACEHOLDER: List subjective variables — thermal preference, noise level, noise kind, earphones, social context (alone/group), activity — and their response categories.]

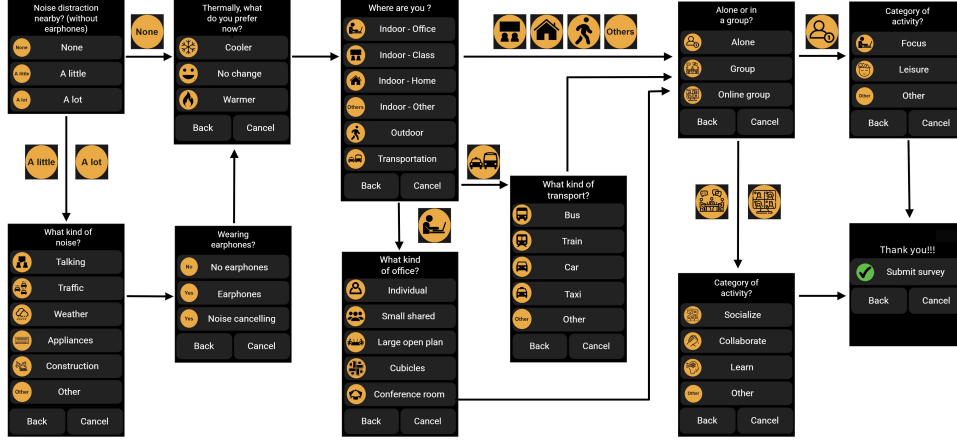


Figure 2. [PLACEHOLDER: Cozie smartwatch micro-survey interface and response flow.]

2.2. Data processing and characterization

[PLACEHOLDER: Describe the EDA strategy from notebook 01 — univariate survey / health distributions, thermal relationships, temporal profiling in Singapore local time, and geospatial mapping.]

[PLACEHOLDER: Describe the characterization methods from notebook 03 — perception fingerprints (row-normalized conditional proportions and lift), association and effect size (Kruskal–Wallis ε^2 , chi-square / Cramér’s V, Spearman ρ), and multivariate structure (MCA + k-modes clustering).]

[PLACEHOLDER: Describe how weather features augment the characterization and the direct test of whether thermal preference tracks actual heat index / rain.]

[PLACEHOLDER: Describe the space-type classifier (features = perceptions $\pm$ weather), validation, and importance / SHAP attribution.]

3. Results

3.1. Data collection, coverage, and temporal characteristics

[PLACEHOLDER: Describe how the study data were collected across Singapore, including geographic coverage, response volume, participant engagement, and the temporal distribution of responses. Explain the weekly and space-type-specific timing patterns that establish the context for the analyses that follow.]

3.2. Survey responses and linked temporal data coverage

[PLACEHOLDER: Summarize the survey battery, response coverage, and the main conditional paths through the questionnaire. Describe the coverage and distributions of linked wearable-health and nearest-station weather observations across the pre-survey aggregation windows.]

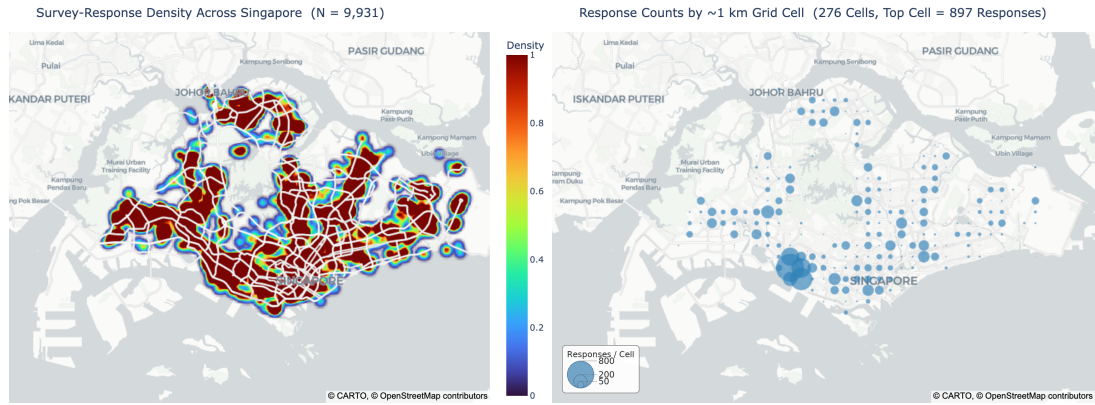


Figure 3. [PLACEHOLDER: Survey-response density and response counts by approximately 1-km grid cell across Singapore (Figure B4).]

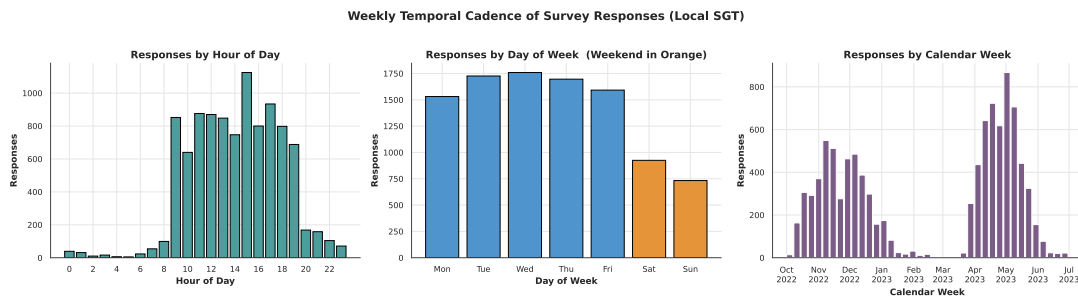


Figure 4. [PLACEHOLDER: Weekly temporal cadence of survey responses in Singapore local time (Figure D2).]

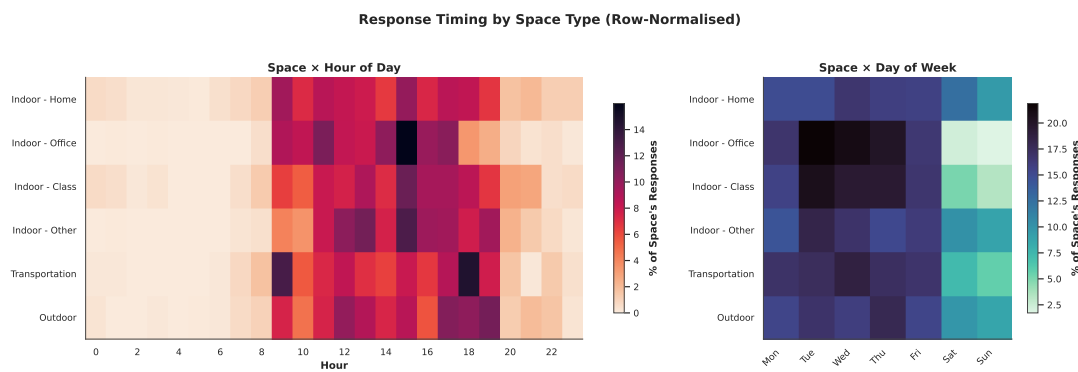


Figure 5. [PLACEHOLDER: Response timing by space type, shown by hour of day and day of week (Figure D4).]

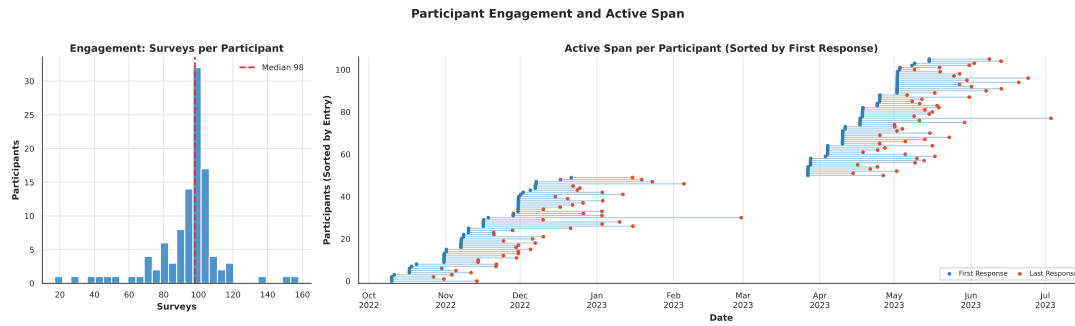


Figure 6. [PLACEHOLDER: Participant engagement and active survey span (Figure E2).]

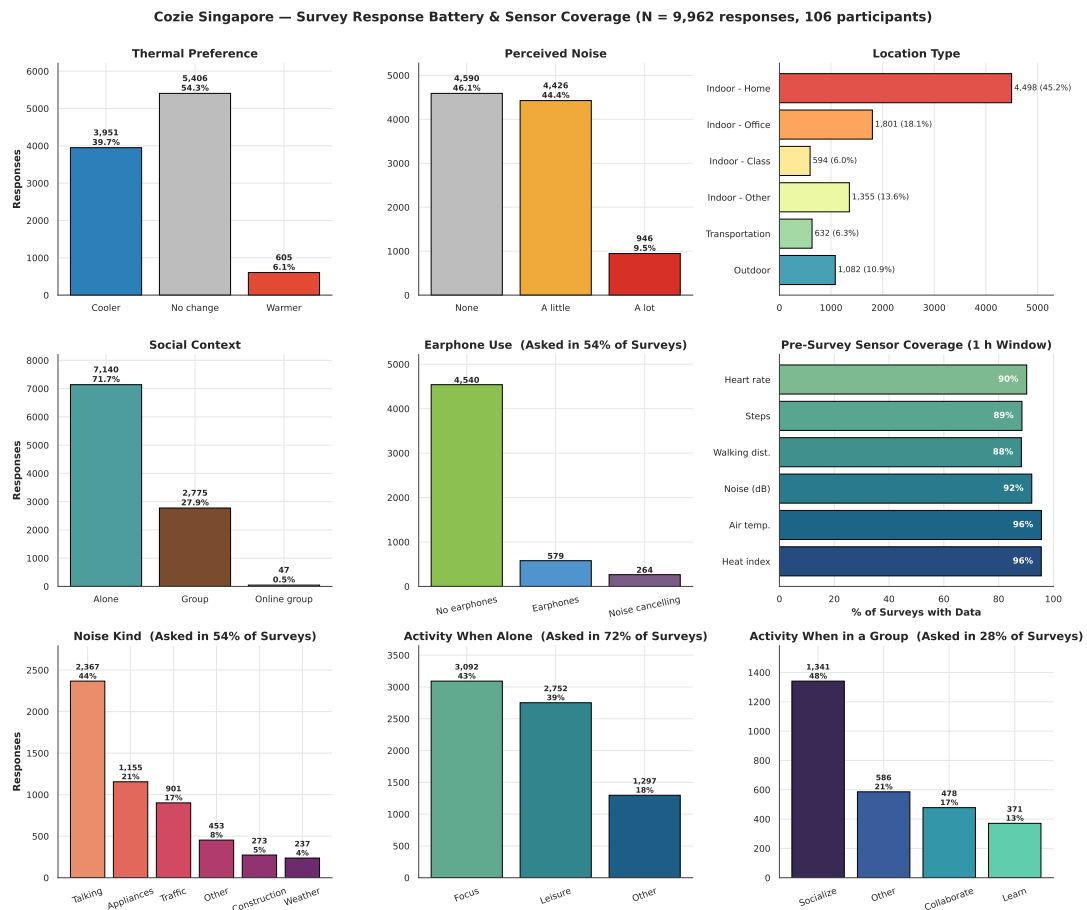


Figure 7. [PLACEHOLDER: Survey response battery, conditional follow-ups, and linked-sensor coverage (Figure A1).]

Survey Response Flow Across the Full Question Battery

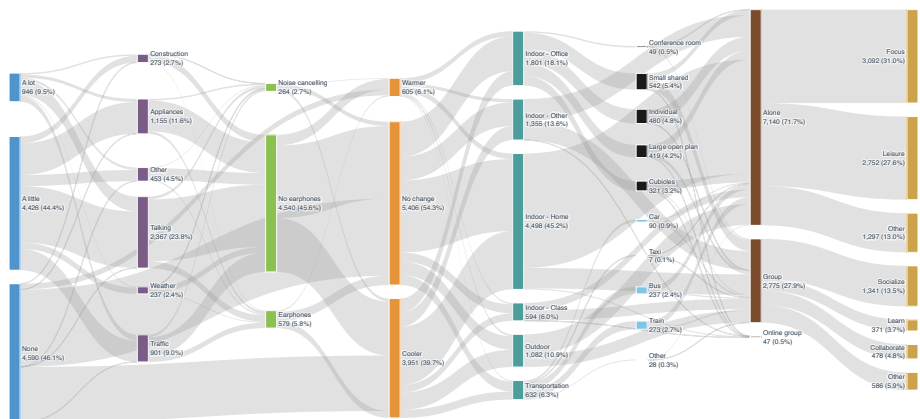


Figure 8. [PLACEHOLDER: Conditional response flow through the watch-survey questionnaire.]

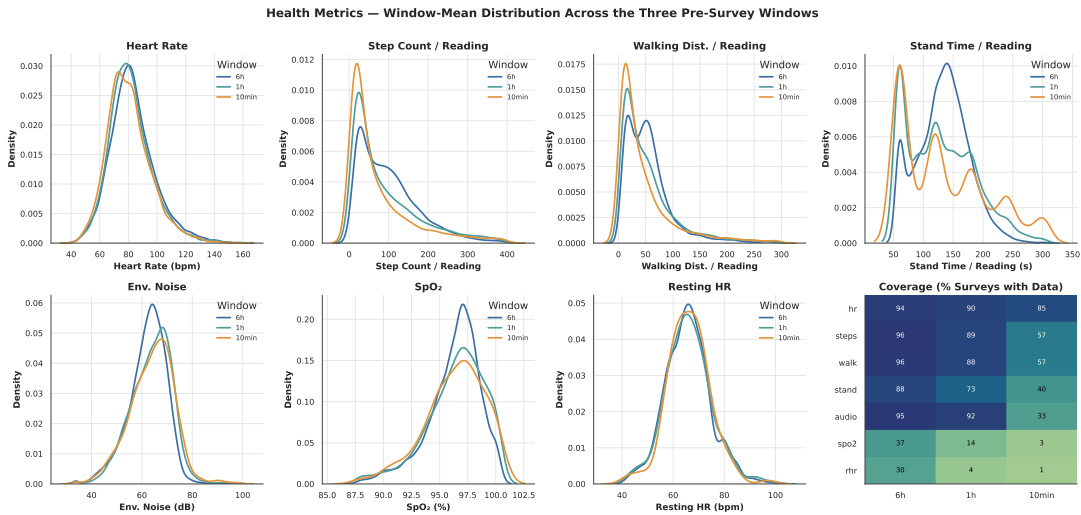


Figure 9. [PLACEHOLDER: Wearable-health metrics across pre-survey aggregation windows (Figure A2).]

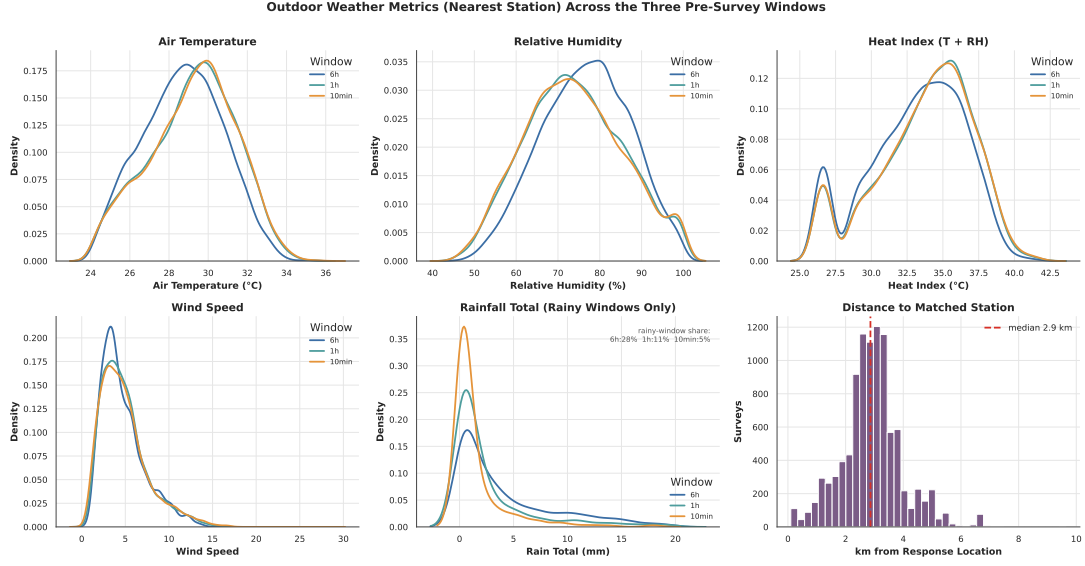


Figure 10. [PLACEHOLDER: Nearest-station outdoor ambient weather across pre-survey aggregation windows (Figure A3).]

3.3. Thermal preference and noise characterization

[PLACEHOLDER: Provide an overview of thermal preference and acoustic experience across participants, contexts, and reported locations. Explain what the cooling-preference and noise-characterization figures establish about the distribution of these outcomes before examining their relationships with space type.]

3.4. Perceptual fingerprints and association with space type

[PLACEHOLDER: Explain the conditional perception profiles in Figure 13. Identify the dominant response patterns within each space type across thermal preference, noise, earphone use, social context, and activity. Make clear that each row is normalized within a space type, so values describe the composition of experiences reported in that space rather than absolute response volume.]

[PLACEHOLDER: Explain the relative distinctiveness in Figure 14. Highlight response patterns that are over-represented or under-represented relative to the full survey corpus, and distinguish these relative contrasts from the raw within-space proportions in Figure 13.]

[PLACEHOLDER: Interpret the cell-level contrasts in Figure 15. Identify the space-response combinations that occur more or less often than expected under independence, focus on coherent patterns across related variables, and treat the significance markers as descriptive screens because responses are nested within participants.]

3.5. Multivariate, physiological, weather, and outdoor environmental patterns

[PLACEHOLDER: Integrate the multivariate structure of subjective experience (MCA + k-modes) with physiological and weather signatures across space types. Then describe the outdoor-only environmental characterization, including streetscape and

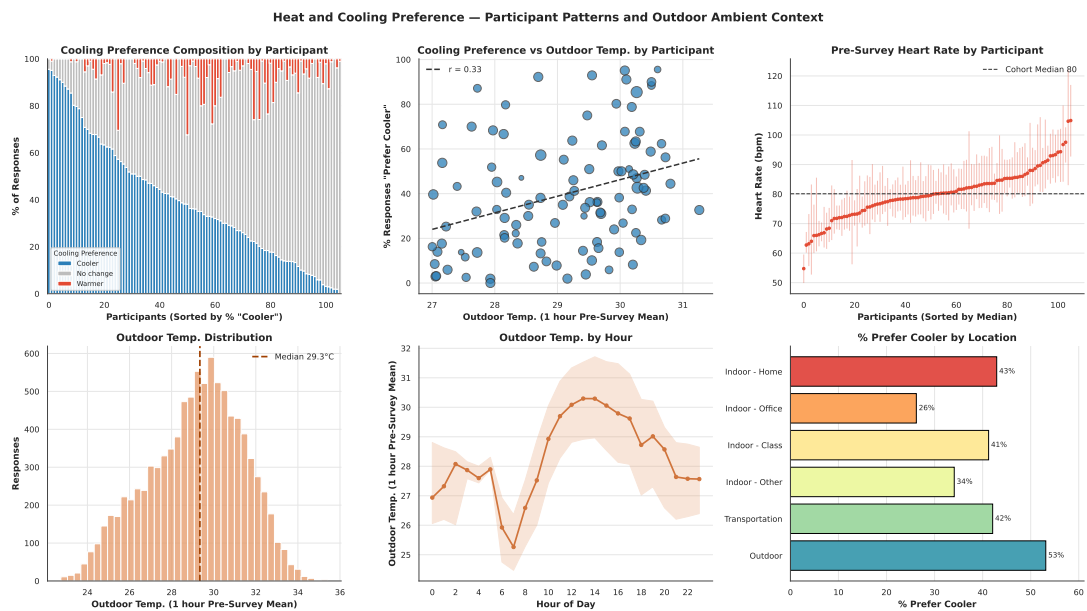


Figure 11. [PLACEHOLDER: Participant-level and contextual patterns in cooling preference (Figure F2).]

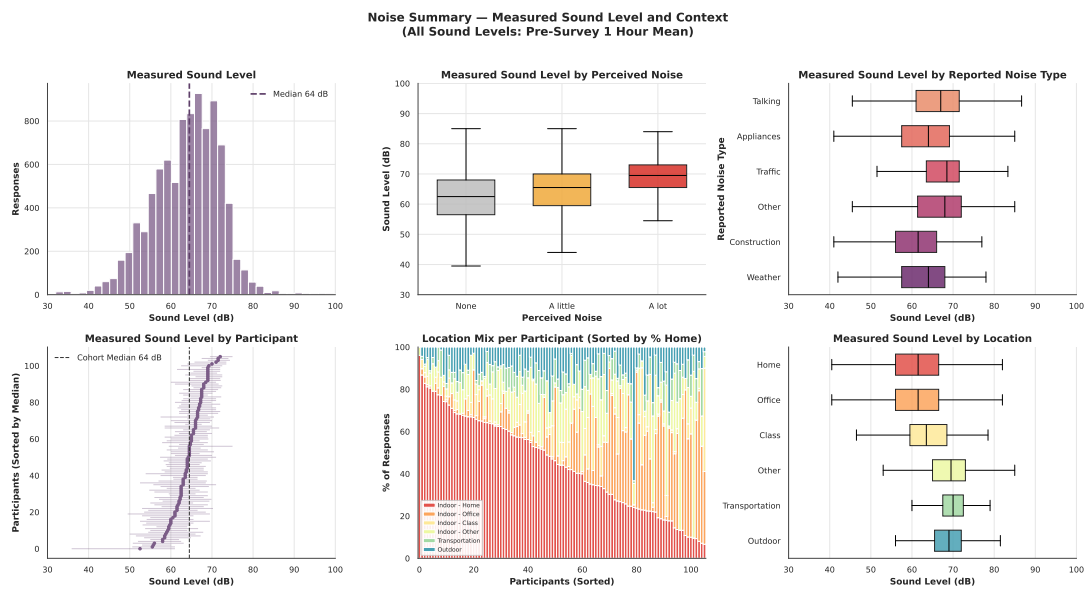


Figure 12. [PLACEHOLDER: Measured sound level across subjective, participant, and location contexts (Figure G1).]

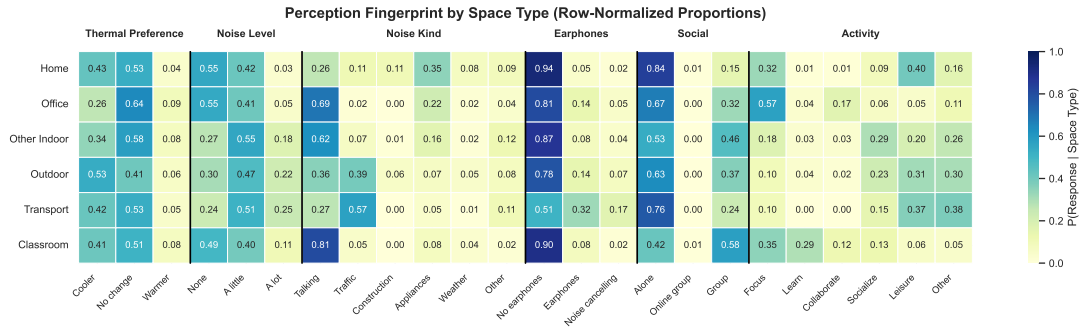


Figure 13. [PLACEHOLDER: Conditional perception fingerprint by reported space type (Figure A1). Cells show $P(\text{response} \mid \text{space type})$.]

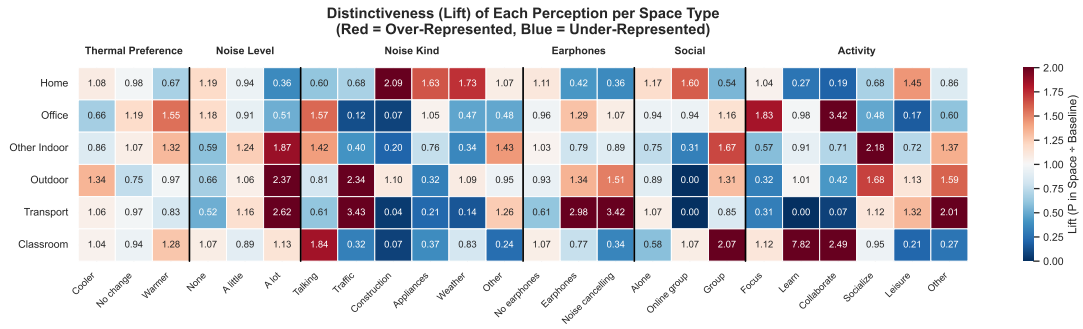


Figure 14. [PLACEHOLDER: Relative distinctiveness of perceptions by reported space type (Figure A2). Cells show lift, $P(\text{response} \mid \text{space type})/P(\text{response})$.]

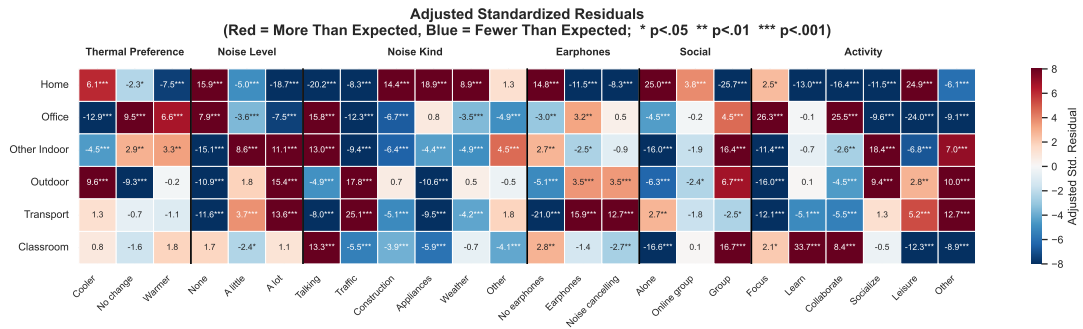


Figure 15. [PLACEHOLDER: Adjusted standardised residuals for space type and perceptual-response combinations (Figure B2). Red cells occur more often than expected under independence and blue cells occur less often than expected.]

built-form profiles, their relationship with thermal preference, and the role of weather.]

[PLACEHOLDER: Report the physiological gradient across spaces (sedentary Home to mobile Outdoor/Transport; steps largest effect), the weak ambient-weather signatures of space type, and the result that Outdoor is not necessarily the hottest reported space.]

[PLACEHOLDER: Report the weather-versus-thermal-preference result: “Cooler” responses occur in genuinely hotter (higher heat-index) and drier conditions, while rain suppresses cooling demand. State the magnitude and limitations of this relationship.]

4. Discussion

4.1. Principal findings

[PLACEHOLDER: Synthesize: function/behaviour (activity, noise + coping, movement) separates space types robustly; thermal, environmental and physiological expectations are mostly weak or convergent nulls; the weather correction shows ambient weather is nearly useless for identifying a space yet genuinely drives outdoor thermal preference.]

4.2. Interpretation against intuition

[PLACEHOLDER: Discuss the reinforced vs contradicted intuitions (e.g. Outdoor is not the hottest; streetscape greenery does not move thermal preference; actual heat does).]

4.3. Implications for sustainable urban design

[PLACEHOLDER: Connect findings to the journal scope — what they mean for thermal-comfort-aware, health-supportive and energy-conscious urban design and policy in tropical cities.]

4.4. Limitations

[PLACEHOLDER: Participant nesting / pseudoreplication; ambient nearest-station weather as a coarse proxy for personal exposure; convenience sample; Singapore-specific A/C context; self-reported perceptions.]

4.5. Future work

[PLACEHOLDER: Directions — personal microclimate sensing, larger/more diverse cohorts, cross-city replication, mixed-effects modeling accounting for participant nesting.]

5. Conclusion

[PLACEHOLDER: Concise restatement of the contribution and headline takeaway for sustainable urban development.]

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CRediT authorship contribution statement

Clayton Miller: Visualization, Software, Resources, Methodology, Funding acquisition, Formal analysis, Writing – original draft, Supervision, Project administration, Investigation, Conceptualization. Yun Xuan Chua: Methodology, Conceptualization, Writing – review & editing, Project administration, Investigation. Mario Frei: Writing – review & editing, Validation, Project administration, Investigation, Conceptualization, Methodology, Data curation.

Declaration of generative AI use

The authors used Runcell AI, powered by OpenAI’s GPT-5.6 Terra large language model, and Claude Opus 4.8 by Anthropic to assist with the organisation of the L^AT_EX manuscript, figure placement, and language polishing. These tools were not used to generate the study data, conduct the analysis, or replace the authors’ scholarly judgement. All generated suggestions were reviewed and revised by the authors, who take full responsibility for the originality, accuracy, integrity, citations, and final content of this manuscript. Generative AI is not listed as an author.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data and code supporting the findings of this study are available on GitHub.

References

- Abboushi, B., Safranek, S., Rodriguez-Feo Bermudez, E., Pratoomratana, S., Chen, Y., Poplawski, M., & Davis, R. (2022). A review of the use of wearables in indoor environmental quality studies and an evaluation of data accessibility from a wearable device. *Frontiers in Built Environment*, 8.
- Appel-Meulenbroek, R., Steps, S., Wenmaekers, R., & Arentze, T. (2020, January). Coping strategies and perceived productivity in open-plan offices with noise problems. *Journal of Managerial Psychology*, 36(4), 400–414.
- Barbour, E., Davila, C. C., Gupta, S., Reinhart, C., Kaur, J., & González, M. C. (2019, August). Planning for sustainable cities by estimating building occupancy with mobile phones. *Nature Communications*, 10(1), 3736.
- Castaldo, V. L., Pigliautile, I., Rosso, F., Cotana, F., De Giorgio, F., & Pisello, A. L. (2018, November). How subjective and non-physical parameters affect occupants’ environmental comfort perception. *Energy and Buildings*, 178, 107–129.
- Christensen, T. C., Barrett, L. F., Bliss-Moreau, E., & Lebo, K. (2003, March). A practical guide to experience-sampling procedures. *Journal of Happiness Studies*, 4(1), 53–78.
- Fischer, T., Schraivogel, S., Caversaccio, M., & Wimmer, W. (2022, March). Are smartwatches a suitable tool to monitor noise exposure for public health awareness and otoprotection? *Frontiers in Neurology*, 13, 856219.
- Fuller, D., Shareck, M., & Stanley, K. (2017, October). Ethical implications of location and accelerometer measurement in health research studies with mobile sensing devices. *Social Science & Medicine*, 191, 84–88.
- Gharani, P., Karimi, H. A., Syzdykbayev, M., Burke, L. E., Rathbun, S. L., Davis, E. M., ... Mendez, D. D. (2021, June). Geographically-explicit ecological momentary assessment (GEMA) architecture and components: Lessons learned from PMOMS. *Informatics for Health and Social Care*, 46(2), 158–177.
- Hammoud, R., Tognin, S., Smythe, M., Gibbons, J., Davidson, N., Bakolis, I., & Mechelli, A. (2024, April). Smartphone-based ecological momentary assessment reveals an incremental association between natural diversity and mental wellbeing. *Scientific Reports*, 14(1), 7051.
- Ignatius, M., Lim, J., Gottkehaskamp, B., Fujiwara, K., Miller, C., & Biljecki, F. (2024, June). Digital twin and wearables unveiling pedestrian comfort dynamics and walkability in cities. In *Isprs annals of the photogrammetry, remote sensing and spatial information sciences* (Vol. X-4/W5-2024, pp. 195–202).
- Ito, K., Kang, Y., Zhang, Y., Zhang, F., & Biljecki, F. (2024, September). Understanding urban perception with visual data: A systematic review. *Cities*, 152, 105169.
- Kingsbury, C., Buzzzi, M., Chaix, B., Kanning, M., Khezri, S., Kiani, B., ... Kestens, Y. (2024, June). STROBE-GEMA: A STROBE extension for reporting of geographically explicit ecological momentary assessment studies. *Archives of Public Health*, 82(1), 84.
- Kirchner, T. R., & Shiffman, S. (2016, September). Spatio-temporal determinants of mental health and well-being: Advances in geographically-explicit ecological momentary assessment (GEMA). *Social Psychiatry and Psychiatric Epidemiology*, 51(9), 1211–1223.
- Miller, C., Abdelrahman, M., Chong, A., Biljecki, F., Quintana, M., Frei, M., ... Wong, D. (2021, November). The internet-of-buildings (IoB): Digital twin convergence of wearable and IoT data with GIS/BIM. *Journal of Physics: Conference Series*, 2042(1), 012041.
- Miller, C., Chua, Y. X., Quintana, M., Lei, B., Biljecki, F., & Frei, M. (2025, October). Make yourself comfortable: Nudging urban heat and noise mitigation with smartwatch-based just-in-time adaptive interventions (JITAI). *Building and Environment*, 284, 113388.
- Miller, C., Quintana, M., Frei, M., Chua, Y. X., Fu, C., Picchetti, B., ... Biljecki, F. (2023, November). Introducing the cool, quiet city competition: Predicting smartwatch-reported heat and noise with digital twin metrics. In *Proceedings of the 10th ACM international conference on systems for energy-efficient buildings, cities, and transportation* (pp. 298–299). New York, NY, USA: Association for Computing Machinery.

- Reichert, M., Giurgiu, M., Koch, E., Wieland, L. M., Lautenbach, S., Neubauer, A. B., ... Liao, Y. (2020, September). Ambulatory assessment for physical activity research: State of the science, best practices and future directions. *Psychology of Sport and Exercise*, 50, 101742.
- Schweiker, M., Risetto, R., & Wagner, A. (2020, March). Thermal expectation: Influencing factors and its effect on thermal perception. *Energy and Buildings*, 210, 109729.
- Stone, A. A., & Shiffman, S. (1994, September). Ecological momentary assessment (EMA) in behavioral medicine. *Annals of Behavioral Medicine*, 16(3), 199–202.
- Tartarini, F., Frei, M., Schiavon, S., Chua, Y. X., & Miller, C. (2023, November). Cozie apple: An iOS mobile and smartwatch application for environmental quality satisfaction and physiological data collection. *Journal of Physics: Conference Series*, 2600(14), 142003.
- Wang, Z., de Dear, R., Luo, M., Lin, B., He, Y., Ghahramani, A., & Zhu, Y. (2018, June). Individual difference in thermal comfort: A literature review. *Building and Environment*, 138, 181–193.
- Yap, W., Stouffs, R., & Biljecki, F. (2023, July). Urbanity: Automated modelling and analysis of multidimensional networks in cities. *npj Urban Sustainability*, 3(1), 1–11.
- Zhang, X., Zhou, S., Kwan, M.-P., Su, L., & Lu, J. (2020, November). Geographic ecological momentary assessment (GEMA) of environmental noise annoyance: The influence of activity context and the daily acoustic environment. *International Journal of Health Geographics*, 19(1), 50.